Fiscal Year:	FY 2008	**Task Last Updated:**	FY 07/25/2007
PI Name:	Chen, David J. Ph.D.		
Project Title:	DNA Damage Responses Induced By HZE Particles In Human Cells		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation Biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	david.chen@utsouthwestern.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	214-648-5595
Organization Name:	University of Texas Southwestern Medical Center		
PI Address 1:	Radiation Oncology, Molecular Radiation Biology		
PI Address 2:	5323 Harry Hines Boulevard		
PI Web Page:			
City:	Dallas	**State:**	TX
Zip Code:	75390	**Congressional District:**	30
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Radiation Biology and Shielding 03-OBPR-07
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No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Changed Division and Discipline/Program to HRP as of FY2006, per program changes at that time, per JSC/A. Chu-ARC (jvp 6/2009) NOTE: Received NCE to 9/30/2009 per PI (7/2008)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):			
Grant/Contract No.:	NNA05CM04G		
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Task Description:	In this project we propose to verify the hypothesis that high energy HZE particles would generate a different and/or unique radiation response in mammalian cells as compared to X-rays. The primary objective of this project is to verify the hypothesis that high energy HZE particles would generate a different and/or unique radiation response in mammalian cells as compared to X-rays. Further more, we will verify whether HZE radiation would induce a bystander effect. Towards this end we will utilize recent advances in our knowledge of molecular mechanisms of DNA damage sensing, signaling, and repair, and changes in global gene expression patterns. Our specific aims are: 1) To test the hypothesis that HZE particles may induce a different gene expression profile than that induced by low LET radiation; 2) To verify the hypothesis that DNA double-strand breaks induced by HZE particles are persistently unrejoined in normal human fibroblasts; 3) To test the hypothesis that HZE particles would induce a persistent and/or altered molecular and cellular response than that induced by X-rays in mammalian cells; 4) To test the hypothesis that high HZE may generate factors that induce a bystander effect or bystander-like effect. We plan to use DNA-damage responses as end points in combination with a novel track-detecting device to fulfill these objectives. The proposed research should reveal the fundamental differences in cellular responses between low LET X-ray and HZE exposures in humans. These studies would, in turn, provide biological markers for risk assessment and guidance in radiation protection for astronauts during space missions in the future. Specific Objectives A.1. To test the hypothesis that HZE particles may induce a different gene expression profile than that induced by low LET radiation. 1) To determine global gene expression patterns induced by HZE and X-rays at a dose of equal survival level. Metabolic pathways and expression markers specific to HZE responses in normal human fibroblasts will be identified. Especially, we will focus on pathways/gene groups related to cell proliferation, stress responses, DNA repair and protein kinases. 2) To determine whether gene expression profiles induced by HZE will vary as a function of energy of the HZE particles. 3) To determine whether transfer of media from HZE-irradiated cells to non-irradiated cells would induce specific gene expression changes in the recipient cells. A.2. To verify the hypothesis that DNA double-strand breaks induced by HZE particles are persistently unrejoined in normal human fibroblasts. 1) To determine the RBE of gamma-H2AX phosphorylation and focus formation by HZE and X-rays at equal survival doses in normal human fibroblasts. 2) To determine whether gamma-H2AX phosphorylation is significantly more persistent at the sites of HZE-induced DSBs as compared to X-rays. 3) To determine whether the induction and persistence of H2AX phosphorylation and focus formation will vary as a function of energy of the HZE particles. A.3. To test the hypothesis that HZE particles would induce a persistent and/or altered molecular and cellular response than that induced by X-rays in mammalian cells. 1) To determine whether key DNA DSB-sensing protein kinases, DNA-dependent protein kinase (DNA-PKcs) and ataxia telangiectasia mutated (ATM), are persistently phosphorylated and activated upon HZE exposure in human fibroblasts. 2) To determine whether the key DSB-signaling and damage-induced cell cycle check point proteins (the MRN complex, MDC1, SMC1, Chk1, Chk2, and p53) are persistently localized on DNA damage sites and/or phosphorylated upon HZE particle exposure in human fibroblasts. 3) To determine if the interaction between homologous recombination (HR) and nonhomologous end joining (NHEJ) proteins with DNA lesions induced by HZE is different from that observed with X-rays. A.4. To test the hypothesis that high HZE may generate factors that induce a bystander effect or bystander-like effect. 1) To determine the extent of DSB induction mediated by bystander and bystander-like effects in HZE-irradiated cells, as measured by H2AX phosphorylation and focus formation and localization of DSB sensing and signaling proteins. 2) To determine whether bystander effects and bystander-like effects are mediated by ROS or modulated through lipid peroxidation in HZE-particle irradiated cells. 3) To determine whether DSB repair capability (HR and/or NHEJ) plays a role in modulating HZE particles-induced bystander and bystander-like effects in mammalian cells.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research will provide biological markers for risk assessment and guidance in radiation protection for astronauts during space missions in the future.
Task Progress:	1. The ability of cells to repair DSBs is clearly dependent on the LET of the radiation. 2. Exposure of monolayer cells in parallel to the ion path leads to the formation of H2AX, 53BP1 and DNA-PKcs foci and their foci dissolution kinetics is similar to that gamma H2AX. 3. Spatial colocalization studies showed that the phosphorylated DNA-PKcs covers only part of the total area of gamma H2AX and 53BP1 foci, thus representing only the DSB site and can serve as a surrogate marker for examining the frank DNA DSB present within the clustered lesion. 4. Werner syndrome protein is recruited to the sites of DNA breaks generated by Fe-particles and in its absence cells become sensitive to Fe-particles. Overall, our current data clearly indicates the fundamental differences in cellular responses to low- and high-LET radiation exposures and that the ability of a cell to repair DSBs is clearly dependent on the LET of the radiation impinging upon it. Our data further support the idea that high-LET induced DNA damage is more difficult to repair and as a consequence, higher biological effectiveness is observed for HZE particles.
Bibliography Type:	Description: (Last Updated: 08/14/2018)
Abstracts for Journals and Proceedings	Asaithamby A, Uematsu N, Chen DJ, "Visualization of DNA double strand breaks repair in living cells." Presented at the NASA Investigators Workshop, Sonoma, CA, July 13-15, 2007. Abstracts, NASA Investigators Workshop, 2007. , Jul-2007
Abstracts for Journals and Proceedings	Asaithamby A, Yannone SM, Chen DJ. "Werner Syndrome Protein is phosphorylated by DNA-dependent protein kinase catalytic subunit and regulates DNA double-strand break repair." Presented at the International Congress on Radiation Research, San Francisco, CA, July 8-12, 2007. Abstracts, International Congress on Radiation Research, 2007. , Jul-2007

Abstracts for Journals and Proceedings	Uematsu N, Ding L, Burma S, Story MD, Chen B, Chen DJ. "Repair kinetics of DNA damage induced by high energy charged particles and gamma-rays." NASA Models of Space Radiation Risks Workshop, at the University of Texas Southwestern Medical Center, Dallas, Texas, March 3-7, 2007. Abstracts, NASA Models of Space Radiation Risks Workshop, 2007. , Mar-2007
Abstracts for Journals and Proceedings	Uematsu N, Asaithamby A, Burkhard J, Taucher-Scholz G, Chen DJ. "Dynamics of DNA repair proteins at DNA damage sites in living cells." NASA Models of Space Radiation Risks Workshop, at the University of Texas Southwestern Medical Center, Dallas, Texas, March 3-7, 2007. Oral Presentation (by DJC), Abstracts, NASA Models of Space Radiation Risks Workshop, 2007. , Mar-2007
Articles in Peer-reviewed Journals	Asaithamby A, Yannone SM, Yano K., Qin J, Chen DJ. "Werner Syndrome Protein is phosphorylated by DNA-dependent protein kinase catalytic subunit and regulates DNA double-strand break repair. " Submitted to Molecular and Cellular Biology, July 2007. , Jul-2007
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